

Bayesian calibrated physics-informed neural networks for second-life battery SOH estimation

Rong Zhu^a, Jiawen Hu^{b,e,*}, Weiwen Peng^{c,d}

^a Department of Systems Engineering, City University of Hong Kong, Hong Kong, China

^b School of Aeronautics and Astronautics, University of Electronic Science and Technology of China, Chengdu 611731, China

^c School of Intelligent Systems Engineering, Shenzhen Campus of Sun Yat-Sen University, Shenzhen 518107, China

^d Guangdong Provincial Key Laboratory of Fire Science and Technology, Guangzhou 510006, China

^e Aircraft Swarm Intelligent Sensing and Cooperative Control Key Laboratory of Sichuan Province, University of Electronic Science and Technology of China, Chengdu, China

ARTICLE INFO

Keywords:

SOH estimation
Second-life batteries
Physics-informed neural networks
Uncertainty quantification
Monte Carlo Dropout

ABSTRACT

Second-life lithium-ion batteries (SL-LIBs) from electric vehicles offer significant potential for extending battery life and reducing environmental impact. Accurate state-of-health (SOH) estimation is critical for ensuring their safe and efficient reuse, but it is challenging due to the heterogeneous aging process and incomplete first-life data. To overcome these challenges, we propose a Bayesian calibrated physics-informed neural network (PINN) framework for SOH estimation, integrating physical degradation models, data-driven calibration techniques and uncertainty quantification. The framework uses simulation data generated by the porous electrode theory-based degradation models to train a governing equation approximator, which serves as a constraint in the PINN for SOH estimation. The Bayesian SOH estimator is trained on real-world data under the guidance of the pretrained governing equation, and leverages Monte Carlo Dropout for uncertainty quantification. Additionally, a physics model calibrator is introduced to bridge the gap between the physics-based model and real-world data, ensuring a more precise representation of actual degradation. Experimental results on a self-designed SL-LIB dataset, consisting of 84 retired batteries across 21 different working conditions, demonstrate that the proposed framework significantly outperforms existing methods.

1. Introduction

The electrification of transportation has become a critical strategy for reducing carbon emissions and tackling climate change [1]. Lithium-ion batteries (LIBs) enable the rapid growth of electric vehicles (EVs) because of their high energy density, low self-discharge rates, and long service life [2]. By 2025, the total installed capacity of EV batteries worldwide is projected to reach 800 GWh, with a market value close to 92 billion dollars [3]. As EV adoption grows, many batteries complete their first service life when their capacity drops to about 80% of the nominal level [4]. These end-of-life batteries still hold significant remaining capacity, making them suitable for second-life applications with lower operational demands, such as energy storage systems (ESS), [5–7]. However, ensuring the safety and reliability of such batteries in second-life applications requires accurate state-of-health (SOH) estimation [8].

Substantial efforts have been made to develop SOH estimation methods for first-life LIBs, as reviewed in [9,10]. Traditional model-based approaches rely on equivalent circuit models [11], electrochemical models [12], and stochastic process models [13], often embedding degradation mechanisms like Dakin's method [14] and Arrhenius-based aging laws [15]. These physics-based methods provide interpretability and can capture capacity-fade processes, including the growth of the solid-electrolyte interface and electrode-material loss [12]. However, model-based methods usually involve extensive parameter tuning and can be computational heavy, which reduce their adaptability to different battery chemistries and working conditions. Data-driven methods, such as Gaussian mixture regression [16], Gaussian process regression [17], relevance vector machine [18], and deep neural networks (DNN) [19–21], extract patterns directly from condition monitoring data like voltage, current, and temperature profiles. While data-driven methods depend less on battery-specific

* Corresponding author.

E-mail address: hdi@sytu.edu.cn (J. Hu).

<https://doi.org/10.1016/j.ress.2025.111432>

Received 3 March 2025; Received in revised form 5 June 2025; Accepted 6 July 2025

Available online 7 July 2025

0951-8320/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

parameters, their performance heavily relies on high-quality datasets, effective feature engineering, and representative training conditions.

Although significant advancements have been made in first-life SOH estimation methods, directly applying these methods to second-life lithium-ion batteries (SL-LIBs) faces additional challenges. Retired batteries often have incomplete usage histories and diverse aging pathways, partly due to privacy concerns and the lack of standardized data-sharing protocols. Moreover, in ESS and other second-life applications, batteries often operate with partial cycles, varied middle state-of-charge (mid-SOC), and inconsistent depths of discharge (DOD) [22]. These factors introduce significant variability in retired batteries' performance, making it hard to evaluate their SOH using standard methods. Developing specialized approaches that address these challenges is important for the safe integration of second-life batteries into ESS and other fields.

Data-driven methods for second-life batteries have drawn considerable attention, due to their ability to capture complex degradation trends without relying heavily on physical modeling [22]. For example, Jiang et al. [23] used a Bayesian neural network with uncertainty calibration to handle partial charge-discharge workloads and achieved a mean absolute percentage error (MAPE) of 2.13%. Tao et al. [4] introduced generative learning to mitigate data scarcity by producing synthetic datasets, achieving MAPE below 6% on batteries with different chemistries and usage histories. Though effective, these methods still depend on data availability, and often struggle with generalization. Recent work using electrochemical impedance spectroscopy with dimensionality reduction [24] and resistive remaining capacity combined with internal resistance [25], has further improved SOH estimation, with root mean square errors (RMSE) as low as 1.18% and 6%, respectively. However, these methods need extra experimental measurements, which can limit their applicability in real-world scenarios.

Physics-informed neural networks (PINNs) have emerged as a promising solution to improve SOH estimation by fusing physical principles with data-driven methods [26,27]. By embedding physical laws directly into the training process, PINNs enforce physics-consistent solutions while retaining the flexibility of neural networks. Wen et al. [28] proposed a semi-empirical partial differential equation for battery degradation dynamics, and used a deep hidden physics model to uncover governing equations when prior knowledge was scarce. Wang et al. [29] presented a PINN framework for SOH estimation based on empirical degradation models and state-space equations. They extracted statistical features from short charge-discharge periods and validated the method using data from 387 first-life batteries, achieving MAPE of 0.87%. Hofmann et al. [30] proposed a PINN framework for SOH estimation, combining the efficiency of neural networks with the accuracy of physico-chemical models. Their method achieved high accuracy, outperforming data-driven models with RMSE below 2% on synthetic data and under 3% on real-world test data. Sun et al. [31] introduced a Battery PINN to enhance the accuracy and interpretability of SOH estimation, integrating physics-based constraints with a feedforward neural network. Several hybrid PINNs have also incorporated equivalent circuit models [32], single-particle models [33], and finite element modeling of SEI growth [34] to enhance SOH estimation by enforcing physical constraints within machine learning.

In most health monitoring applications, defining an explicit dynamic physics model is challenging due to the complexity of real-world systems [28]. Existing PINN methods typically simplify the physics models or use various techniques to embed them into data-driven models. In regularized PINN approaches [28,29], the model trains a neural network as a function approximator to learn the degradation dynamics of the battery based on condition monitoring data. However, these method does not inherently incorporate prior physical knowledge of the battery's degradation process, failing to provide additional information from the battery's physical characteristics. While these methods enable the network learns from the available data, they lack the ability to leverage the physics-based prior knowledge that could potentially improve the model's accuracy and generalizability. Additionally, these methods

might overlook important degradation mechanisms that are crucial for managing second-life batteries, where the working conditions are more variable and less predictable than in first-life usage.

To overcome these limitations, we propose a novel Bayesian calibrated PINN framework that incorporates a pre-trained governing equation approximator, specifically designed for second-life batteries. Given the complexity of the governing equations for lithium-ion batteries, directly embedding the complete governing equations within the PINN model is challenging. Therefore, we first generate simulation data from the porous electrode theory (PET)-based battery degradation model driven by complete governing equations [35], representing various real-world battery working conditions. This simulation data is then used to pre-train the PINN model, which mainly used to train the governing equation approximator. After pre-training, the governing equation approximator is frozen and severed as the physics informed constraint for SOH estimation model training. The SOH estimator is trained using real-world data, with a physics-informed constraint incorporated into the model to ensure that the estimations align with known degradation principles. Inspired by recent research on the data-driven model correction strategies [36,37], we introduce a physics model calibrator to address the gap between the idealized physics model and the real-world observations.

To further enhance the accuracy of the SOH estimation, Bayesian inference is employed to quantify uncertainty in the SOH estimation, providing a more robust SOH estimation framework for SL-LIBs. Uncertainty quantification has gained significant attention in battery health prognostics, as reviewed in [38,39], which demonstrate its importance in enhancing predictive performance. Techniques like Monte Carlo (MC) methods [40,41], Gaussian process regression [27, 42] ensemble learning [18,43] are widely used to capture uncertainties. Additionally, Lin et al. [44] introduced a SOH estimation method that combines gated recurrent unit neural networks and a hidden Markov model to address uncertainty quantification. Zhang et al. [45] incorporated quantile distribution of deep features into the deep learning framework to provide SOH values with associated confidence intervals (CIs). The model uses a Wasserstein distance-based quantile Huber loss function for optimization, combining Huber loss and quantile regression loss. Zhang et al. [46] introduced an automatic machine learning-based method that uses bootstrap sampling for uncertainty quantification. The method allows for offline training with distinct battery datasets, followed by online deployment for practical SOH estimation.

Building upon these existing approaches, we integrate MC Dropout [47], a well-established method for uncertainty quantification [23,41], into the PINN framework. MC Dropout is applied during the training and testing process of the calibrated PINN to model the uncertainty. By randomly dropping neurons during training, MC Dropout introduces stochasticity into the network, enabling the model to generate multiple predictions for the same input. This process captures epistemic uncertainty, which reflects uncertainty in the model's knowledge. The integration of MC Dropout allows the model to produce CIs, offering a more robust framework for SOH estimation in SL-LIBs.

The main contributions of this work are summarized as follows:

1. A Bayesian calibrated PINN framework is proposed, which integrates a pre-trained governing equation approximator to improve the accuracy of SOH estimation for SL-LIBs. This approach leverages simulation data to pre-train the governing equation approximator, addressing the challenges of directly embedding complex physics models.
2. The framework introduces a data-driven calibration model that learns discrepancies between the physics-based model and real-world data. This calibration model acts as a bridge, refining the representation of complex degradation mechanisms and filling the gap between the idealized physics model and actual working conditions of SL-LIBs.

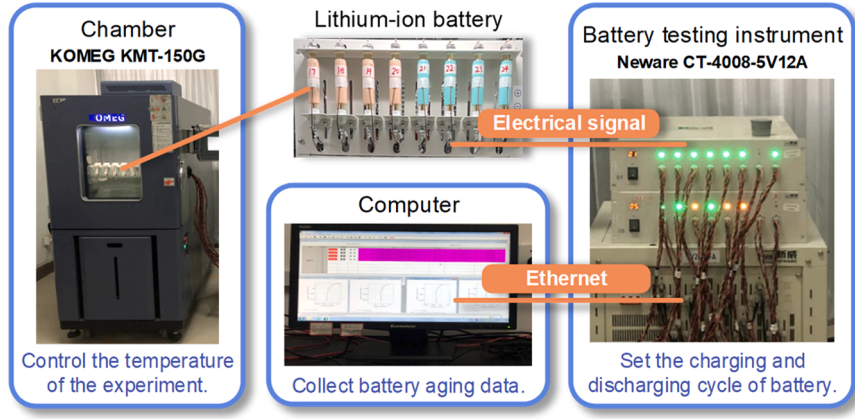


Fig. 1. Overview of the experimental setup used for battery testing.

Table 1

The chemical components and basic experiment conditions.

Basic information	Description
Chemical component	$\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$
Nominal capacity	2600 mAh
Nominal voltage	3.6 V
Cut-off voltage	2.75–4.2 V
Temperature	25°C
Working SOC ranges (Batches)	Batch 1–Batch 6: 70%–90%, 60%–80%, 40%–60%, 30%–50%, 20%–40%, 10%–30% Batch 7–Batch 12: 60%–90%, 45%–75%, 35%–65%, 30%–60%, 50%–90%, 35%–75% Batch 13–Batch 21: 50%–60%, 20%–50%, 10%–50%, 25%–75%, 10%–60%, 20%–80%, 15%–85%, 10%–90%, 0%–100%

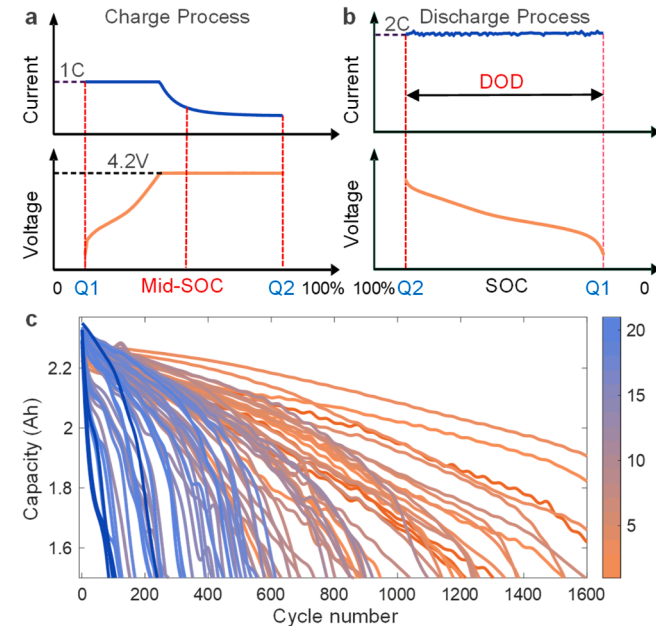


Fig. 2. (a, b) Charging and discharging protocols used in the experiments. (c) Capacity trajectory of 84 retired batteries.

3. To enhance robust SOH estimation, the framework incorporates PINNs with MC Dropout to quantify uncertainties. This addition improves the reliability of SOH estimations, particularly under the variable conditions typical of second-life applications, by providing CIs that reflect the uncertainty in the estimation results.

The remainder of this paper is structured as follows. Section 2 defines the second-life SOH estimation problem and introduces a SL-LIBs dataset. Section 3 details the Bayesian calibrated PINN framework. Section 4 presents a case study using second-life battery dataset to validate the proposed framework. Finally, Section 5 summarizes the findings and proposes directions for future research.

2. Problem statement and data description

2.1. Problem statement

The SOH of LIBs is commonly defined as the ratio of its current usable capacity to the nominal capacity. Let C_r be the nominal capacity of LIBs. After experiencing various working conditions over its first-life service in an electric vehicle, the battery retired with a remaining capacity C_a . The SOH is defined as $\text{SOH} = C_a/C_r \times 100\%$.

Estimating C_a for second-life batteries is more challenging than for new ones. In a second-life scenario, batteries have gone through heterogeneous and often unrecorded aging processes. Furthermore, second-life applications (e.g., ESS) often involve partial cycling with incomplete charge/discharge data and limited measurements. These issues complicate the direct use of both traditional physics-based approaches and purely data-driven methods.

The objective of this paper is to develop a deep learning-based SOH estimation model $f(\cdot)$ that accurately estimates the SOH of SL-LIBs under partial cycling conditions, without access to historical first-life monitoring data. By combining physics-informed methods with data-driven calibration strategies, the proposed approach aims to provide precise SOH estimations.

2.2. Dataset description

In the second-life LIB dataset, a total of 84 second-life lithium-ion batteries, previously used in electric vehicles, underwent cycling aging tests to investigate their degradation behavior. The experimental setups are shown in Fig. 1, and Table 1 summarize the chemical composition and key experimental conditions. The experiments were designed to emulate the diverse working conditions typically observed in ESS applications. To achieve this, the batteries were grouped into 21 batches. The 21 working conditions were selected to cover a broad range of real-world scenarios, with each condition representing a distinct combination of mid-SOC and DOD settings. Four SL-LIBs were allocated to each batch.

For each condition, the batteries were charged using a constant current–constant voltage (CC-CV) protocol. The charging process began at 1C (2.6 A) until the battery voltage reached 4.2 V, followed by a constant voltage stage at 4.2 V until the current dropped to the cutoff

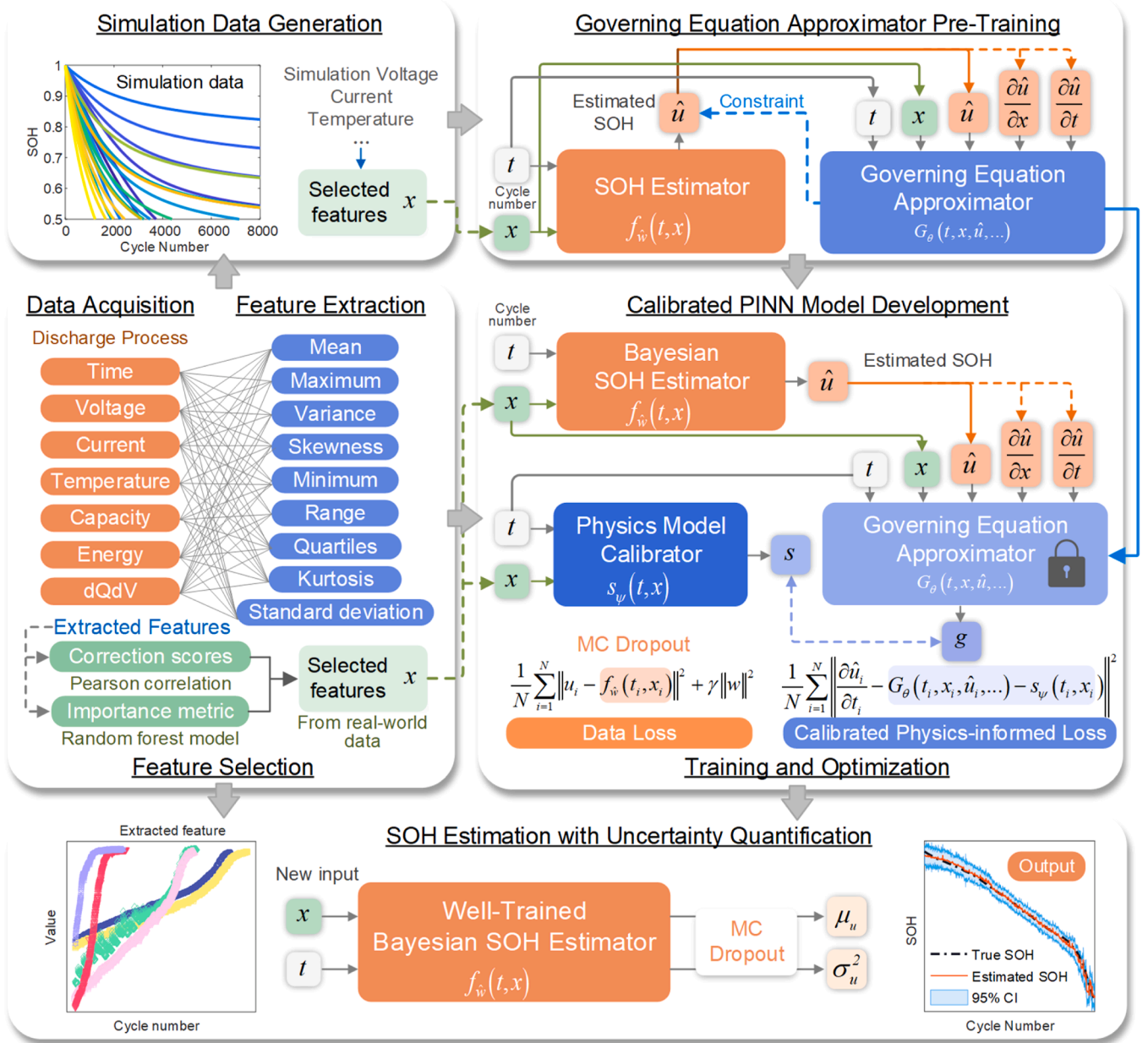


Fig. 3. The proposed Bayesian calibrated PINN framework for SOH estimation.

current. Discharging was performed at 2C (5.2 A) until the voltage reached the predefined DOD constraints, as illustrated in Fig. 2 (a) and (b). Every 25 cycles, a calibration test was conducted to measure the remaining capacity, which provided the ground truth for the SOH. To estimate the SOH for every cycle (not just the calibration points), we applied spline interpolation based on these periodic measurements. The capacity trajectories of all 84 SL-LIBs tested under these conditions are presented in Fig. 2 (c).

3. Bayesian calibrated PINN framework

This section describes the proposed calibrated PINN framework for second-life battery SOH estimation. By integrating PINN, calibration DNN, and uncertainty quantification, this framework provides accurate SOH estimates even under challenging second-life conditions.

3.1. Bayesian calibrated PINN framework for SOH estimation

The proposed calibrated PINN framework aims to improve the accuracy of SOH estimation for SL-LIBs by combining physical degradation models with data-driven learning. Given the complexity of LIB degradation, physics-based models often fail to fully capture real-world performance due to the inherent simplifications and assumptions involved. To address this, the framework first utilizes a PET-based battery degradation model consisting of governing equations to generate simulation data, which provides a realistic representation of battery behavior. This simulation data is used to pre-train the PINN model, allowing the governing equation approximator to learn the general degradation patterns. Once pre-trained, the governing equation approximator is fixed, and the Bayesian SOH estimator is fine-tuned with actual condition monitoring data from SL-LIBs to adjust for the specific conditions and aging mechanisms encountered in these applications.

The overall workflow is illustrated in Fig. 3, which includes the

following steps:

Step 1. Data Acquisition: Data preprocessing begins with collecting raw voltage, current, and temperature data from aging tests conducted under controlled cycling conditions. From these measurements, key curves such as capacity, energy, and incremental capacity are calculated.

Step 2. Feature Extraction: In this step, statistical features are extracted from the computed curves in Step 1. Based on these curves, eleven feature extraction methods such as mean, maximum, minimum, and standard deviation are calculated to represent degradation patterns in battery performance.

Step 3. Feature Selection: Features with the highest relevance to SOH are selected based on Pearson correlation and random forest importance scores, ensuring that only the most significant indicators are used as model inputs.

Step 4. Calibrated PINN Model Development: The model comprises three primary components: 1) a Bayesian SOH estimation model that provide SOH estimation results with uncertainty quantification, 2) a physics-based governing equation approximator that captures the underlying degradation mechanisms of the battery, and 3) a physics model calibrator that learns the discrepancies between physics model and real-world systems. Each component is constructed using fully connected neural networks due to its ability to model the complex, non-linear relationships.

Step 5. Simulation Data Generation: Simulation data is generated based on the PET model to simulate battery degradation under various cycling conditions (e.g., mid-SOC, DOD). This data reflects degradation dynamics such as lithium-ion diffusion and resistance growth, providing a foundation for pre-training the governing equation approximator.

Step 6. Governing Equation Approximator Pre-Training: The governing equation approximator is pre-trained using the simulation data from Step 5. This process enables the model to learn the general degradation patterns of the battery, establishing a baseline understanding of SOH evolution based on the governing equations, which is then used to develop the SOH estimation model with real-world data.

Step 7. Training and Optimization: Following pre-training of the governing equation approximator, the Bayesian SOH estimation model integrated with the physics model calibrator is trained with real-world condition monitoring data, using MC Dropout to capture epistemic uncertainties. A combined loss function balances physical consistency (calibrated physics loss) and alignment with observed data (data loss).

Step 8. SOH Estimation with Uncertainty: The final model delivers SOH estimation results accompanied by CIs, leveraging MC Dropout during inference to provide probabilistic estimations.

3.2. Calibrated PINN structure

The calibrated PINN structure for second-life battery SOH estimation integrates physics-based models with data-driven calibrator. This approach addresses the challenges of model discrepancies and uncertainties in second-life scenarios by combining physical governing equations with a neural network-based physics model calibrator.

(1) PET-based battery degradation model

The PET-based model [35] is employed to model the complex degradation processes in lithium-ion batteries. The model describes the behavior of lithium-ion diffusion within the electrodes using Fick's second law, which governs the concentration gradients of lithium ions in

the solid phase of the electrodes:

$$\frac{\partial}{\partial \tau} c_s(z, \tau) = \frac{1}{z^2} \frac{\partial}{\partial z} \left(z^2 D_s \frac{\partial}{\partial z} c_s(z, \tau) \right). \quad (1)$$

Here, c_s represents the lithium-ion concentration with respect to the spatial variable z and time τ , and D_s is the effective diffusion coefficient. The diffusion process, coupled with the flux of lithium ions at the electrode boundaries, directly impacts the degradation rate, contributing to the growth of internal resistance and capacity fade.

The SOC of the anode is defined by the integral of lithium concentration:

$$SOC(\tau) = \frac{1}{L_n c_{s, \max, n}^s} \int_0^{L_n} c_s(z, \tau) dz, \quad (2)$$

where L_n is the length of the electrode, and $c_{s, \max, n}^s$ is the maximum concentration of lithium ions in the anode. The voltage across the battery is similarly influenced by these concentration dynamics, as described by $V(\tau) = \Phi_s(0, \tau) - \Phi_s(L, \tau)$. Here, Φ_s represents the solid phase potential.

Battery degradation is complex influenced by both electrochemical and mechanical processes. A key aspect of the degradation process is the solid electrolyte interphase (SEI) capacity fade, which reduces the total capacity of the negative electrode and the overall battery due to a parasitic side reaction. The irreversible side reaction, which is central to SEI growth, is defined by the Tafel relationship:

$$j_{side}(z, \tau) = -\frac{i_{0, side}}{F} \left(\frac{I(\tau)}{Q_0^n} \right)^m \exp \left(-\alpha_c \frac{F \eta_{side}(z, \tau)}{RT(z, \tau)} \right), \quad (3)$$

where $i_{0, side}$ is the exchange current density for the side reaction, and α_c is the cathodic transfer coefficient. The side reaction overpotential, $\eta_{side}(z, \tau)$, accounts for the driving force of the side reaction, which depends on the local potential at the electrode and the reference potential voltage. Additionally, R and $T(z, \tau)$ represent the universal gas constant and the local temperature, respectively, which influence the reaction rate.

The SOH of the battery, which reflects the remaining capacity relative to the initial capacity, can be modeled based on the SEI fade model as [35]:

$$SOH(\tau) = 100\% \left(1 - \frac{F a^n}{Q_0^n} \int_0^\tau \int_0^{L_n} j_{side}(z, \tau') dz d\tau' \right). \quad (4)$$

Here, F is Faraday's constant, a^n represents the charge loss due to the side reaction, Q_0^n is the initial capacity of the negative electrode, and $j_{side}(z, \tau')$ is the side reaction current density at a given position z and time τ' . The integral terms reflect the cumulative effect of the SEI layer's growth over both time and spatial coordinates, which decreases the efficiency of the negative electrode, and leads to further degradation of the battery's performance. More details can refer to [35]. Although the physics-based degradation model captures the main degradation mechanisms, it may not account for localized effects such as electrode cracking or dendrite formation. To address this, we incorporate a data-driven calibration module that calibrates the estimation results based on real-world data. This allows the model to adapt to unmodeled degradation behaviors and improve the accuracy of SOH estimation.

(2) Governing equations as the physics foundation

While the PET provides a comprehensive understanding of battery degradation under ideal conditions, the explicit form of the degradation model, represented by $u = f(t, x)$, becomes difficult to obtain. Here, u denotes the SOH, t represents the cycle number, and x is a vector of influencing factors such as mid-SOC, DOD, temperature, and other health-related features. To capture the degradation dynamics, the SOH

decay can be expressed through a governing equation $g(t, x, u; \theta)$, which takes the form of a partial differential equation. This equation g is governed by a nonlinear function of t , x , and u , parameterized by θ that dictates the degradation rate.

For second-life batteries, where real-world conditions differ significantly from ideal assumptions, the explicit form of g is often unknown or too complex to obtain. To handle this, a basic PINN framework incorporates a neural network G_θ to approximate g [29]. The governing equation is rewritten as:

$$\frac{\partial u}{\partial t} \approx G_\theta \left(t, x, u, \frac{\partial u}{\partial t}, \frac{\partial x}{\partial t}, \dots \right). \quad (5)$$

This neural network G_θ integrates the physical constraints derived from the porous electrode theory with data-driven techniques. By approximating the complex function g , the PINN framework adapts to the real-world conditions of second-life batteries, which often exhibit sparse data and variable aging trajectories. This approach ensures that the degradation model remains consistent with known battery physics while being flexible enough to account for the diverse aging patterns observed in second-life battery applications. This strategy maintains consistency with known degradation physics while adapting to second-life battery conditions, which often feature sparse data and varied aging trajectories.

(3) DNN-based calibrator

The governing equations offer a foundation for SOH estimation, but real-world conditions differ significantly from these ideal assumptions. SL-LIBs can exhibit additional effects that are not captured in standard models. Incomplete usage data, different working conditions, and heterogeneous aging histories can all introduce inaccuracies. To address these issues, the calibrated PINN adds a residual calibration term $s_\psi(t, x)$, parameterized by a DNN. This term refines the physics-based model's estimations by accounting for unknown or unmodeled degradation factors. The calibrated governing equation is expressed as:

$$\frac{\partial u}{\partial t} \approx G_\theta \left(t, x, u, \frac{\partial u}{\partial t}, \frac{\partial x}{\partial t}, \dots \right) + s_\psi(t, x). \quad (6)$$

Here, $s_\psi(t, x)$ learns the discrepancies caused by diverse working conditions and complex degradation pathways. This hybrid approach adapts to second-life scenarios while retaining physical consistency.

The residual calibration DNN is a feedforward network that maps input features (t, x) to the calibration term $s_\psi(t, x)$. Mathematically, the output of the DNN is:

$$s_\psi(t, x) = W^{(L)} \sigma(W^{(L-1)} \sigma(\dots \sigma(W^{(1)}(t, x) + b^{(1)}) \dots) + b^{(L-1)}) + b^{(L)}, \quad (7)$$

where $W^{(l)}$ and $b^{(l)}$ are the weights and biases of the l -th layer, $\sigma(\cdot)$ is the activation function, and L is the total number of layers. The network parameters $\psi = \{W^{(l)}, b^{(l)}\}$ are optimized during training to minimize the overall loss function. The calibration process is balanced to ensure both physical consistency and data consistency, ensuring the model aligns with real-world observations while respecting underlying physical laws.

3.3. Uncertainty quantification and loss function

The training process for the calibrated PINN framework uses MC Dropout for uncertainty quantification, and employs a loss function that balances data fidelity, physics consistency, and model calibration.

(1) MC Dropout for uncertainty quantification

In Bayesian neural networks, the posterior distribution $P(w|D)$ over weights w , given data $D = \{(t_i, x_i), u_i\}_{i=1}^N$, is defined by Bayes' theorem:

$$P(w|D) = \frac{P(D|w)P(w)}{P(D)}, \quad (8)$$

where $P(D|w)$ is the likelihood of the data given the parameters w , $P(w)$ is the prior distribution over weights w , and $P(D)$ is the marginal likelihood. Direct computation of the posterior $P(w|D)$ is often intractable for DNN due to the high-dimensional parameter space.

Variational Inference (VI) addresses this by approximating the true posterior $P(w|D)$ with a simpler variational distribution $q_\lambda(w)$, parameterized by λ . The goal is to minimize the Kullback-Leibler (KL) divergence between $q_\lambda(w)$ and the true posterior:

$$KL(q_\lambda(w)P(w|D)) = \int q_\lambda(w) \log \frac{q_\lambda(w)}{P(w|D)} dw. \quad (9)$$

Minimizing this KL divergence is equivalent to maximizing the evidence lower bound (ELBO):

$$ELBO(q_\lambda) = E_{q_\lambda(w)}[\log P(D|w)] - KL(q_\lambda(w)P(w)), \quad (10)$$

The first term increases the likelihood of the data, and the second term constrains the posterior to remain near the prior.

To avoid explicitly parameterizing $q_\lambda(w)$, MC provides a practical approximation of the weight posterior [47]. Rather than modeling $q_\lambda(w)$ directly, MC Dropout introduces Bernoulli-distributed random variables $\varepsilon_i \sim \text{Bernoulli}(p_i)$ for each layer i . These random variables deactivate a fraction of the weights, effectively sampling different regions of the parameter space in each forward pass. Each forward pass with dropout samples a different w from $q_\lambda(w)$.

During testing, performing T forward passes with dropout activated produces samples $\{\hat{w}_t\}_{t=1}^T$. For a new input x^* , the estimations $\hat{u}_t = f_{\hat{w}_t}(x^*)$ approximate the posterior predictive distribution:

$$P(\hat{u}|x^*, D) \approx \sum_{t=1}^T p(\hat{u}|x^*, \hat{w}_t) / T. \quad (11)$$

The predictive mean μ_u and variance σ_u^2 across these T samples provide a central estimate and uncertainty measure, which are calculated as:

$$\mu_u = \sum_{t=1}^T \hat{u}_t / T, \quad \sigma_u^2 = \sum_{t=1}^T (\hat{u}_t - \mu_u)^2 / T. \quad (12)$$

(2) Loss function design

The total loss function integrates data fidelity and calibrated physics constraints, achieving a balance between accurate estimations, physics consistency, and robust uncertainty quantification.

For the data fidelity term L_{data} , consider a set of observed input-output pairs $D = \{(t_i, x_i), u_i\}_{i=1}^N$. Under a Gaussian likelihood assumption, maximizing $E_{q_\lambda(w)}[\log P(D|w)]$ is equivalent to minimizing the mean squared error (MSE) between estimations and observations:

$$L_{data} = \sum_{i=1}^N (u_i - f_{\hat{w}}(t_i, x_i))^2 / N + \gamma w^2. \quad (13)$$

Here, $f_{\hat{w}}(t_i, x_i) = \hat{u}_i$ is the estimated output using a dropout-sampled parameter set $\hat{w}$. The parameter γ is the weight decay of each parameter for the ℓ^2 weight regularization.

The calibrated physics-informed term $L_{Cphysics}$ enforces consistency with the governing equations:

$$L_{Cphysics} = \sum_{i=1}^N \left(\frac{\partial u_i}{\partial t_i} - G_\theta \left(t_i, x_i, u_i, \frac{\partial u_i}{\partial t_i}, \frac{\partial x_i}{\partial t_i}, \dots \right) - s_\psi(t_i, x_i) \right)^2 / N. \quad (14)$$

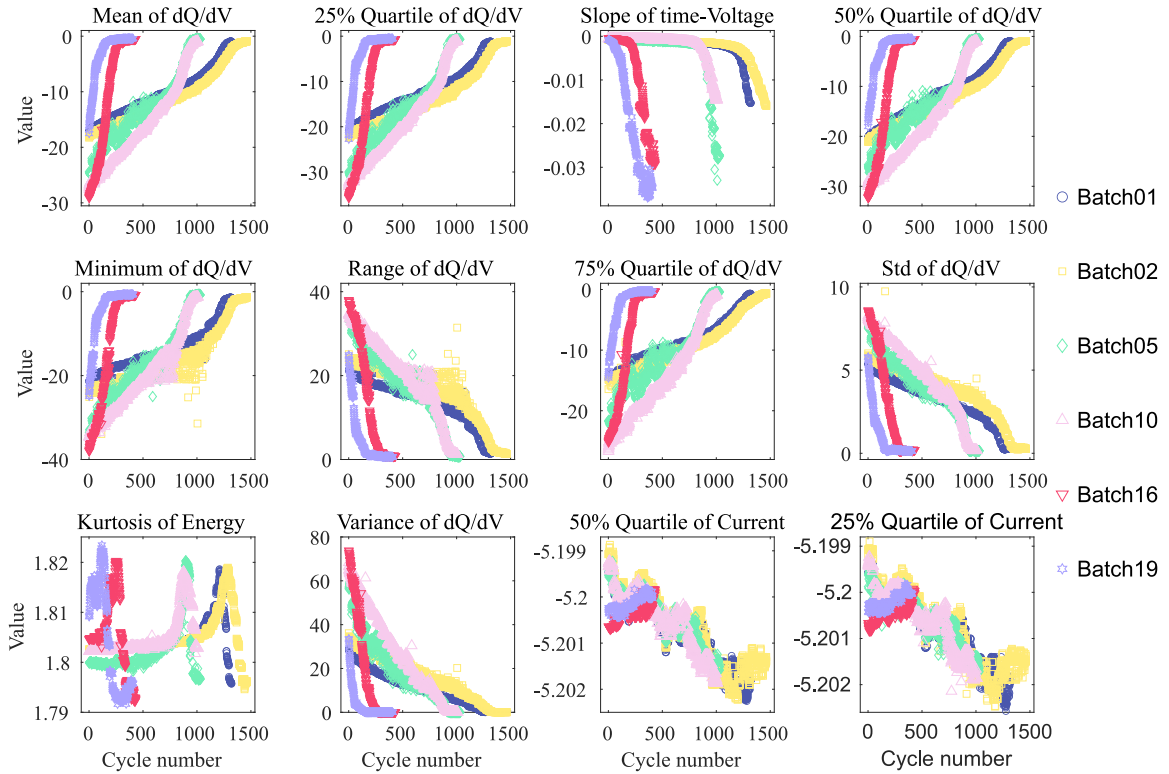
Here, N is the number of data points, $G_\theta(t_i, x_i, u_i, \dots)$ represents the

Algorithm 1

Training and optimization of calibrated PINN framework.

Input: Training data $D = \{(t_i, x_i), u_i\}_{i=1}^N$, governing equations $g(t, x, u; \theta)$, dropout rate p , number of MC samples T **Output:** Trained PINN model with predictive mean and uncertainty estimates

1. Generate simulation data based on PET-based battery degradation model to create training set for the governing equation approximator $D_{sim} = \{(t_i, x_i), u_{sim}\}_{i=1}^M, u_{sim} = g(t_i, x_i, u_i; \theta)$
2. Pre-train the governing equation approximator using simulation data D_{sim} , and obtain the trained governing equation approximator $\hat{G}_\theta$
3. Initialize network parameters w (for SOH estimator) and ψ (for residual calibration DNN)
4. Set learning rate α , regularization coefficient γ , balanced parameter β , and number of epochs E
5. For epoch in 1 to E do:
6. For each batch in D do:
7. Compute data loss: $L_{data} = \sum_{i=1}^N (u_i - f_w(t_i, x_i))^2 / N + \gamma w^2$
8. Compute residuals $R = \partial u / \partial t - \hat{G}_\theta(t, x, u, \dots) - s_\psi(t, x)$
9. Compute calibrated physics-informed loss: $L_{Cphysics} = \sum_{i=1}^N R^2 / N$
10. Compute total loss: $L_{total} = L_{data} + \beta L_{Cphysics}$
11. Compute gradients ∇L_{total} with respect to w and ψ
12. Update parameters $w \leftarrow w - \alpha \nabla L_{total}(w)$
13. Update parameters $\psi \leftarrow \psi - \alpha \nabla L_{total}(\psi)$
14. End For
15. End For
16. # Predictive output with uncertainty
17. For each test sample X_{test} do:
18. Perform T forward passes with dropout, $\hat{u}_t = f(X_{test}; w_t, \theta, \psi)$
19. Compute predictive mean: $\mu_u = \sum_{t=1}^T \hat{u}_t / T$
20. Compute predictive variance: $\sigma_u^2 = \sum_{t=1}^T (\hat{u}_t - \mu_u)^2 / T$
21. Return μ_u and σ_u^2 as the prediction and uncertainty
22. End For

**Fig. 4.** Visualization of top 12 selected features based on importance and correlation.

physics-based component, and $s_\psi(t, x_i)$ is the residual calibration term.

Combining the data loss with the physics-informed loss, the total loss function is:

$$L_{total} = L_{data} + \beta L_{Cphysics} \quad (15)$$

where β balances the contributions of the data fidelity term and calibrated physics-based term. By integrating MC Dropout for uncertainty quantification and combining data fidelity with calibrated physics-

informed constraints, the calibrated PINN framework ensures robust SOH estimation.

3.4. Training and optimization

The training process for the calibrated PINN framework integrates uncertainty quantification, data fidelity, and physics consistency into a unified optimization scheme. The process begins with the generation of simulation data based on the governing equations, which are derived

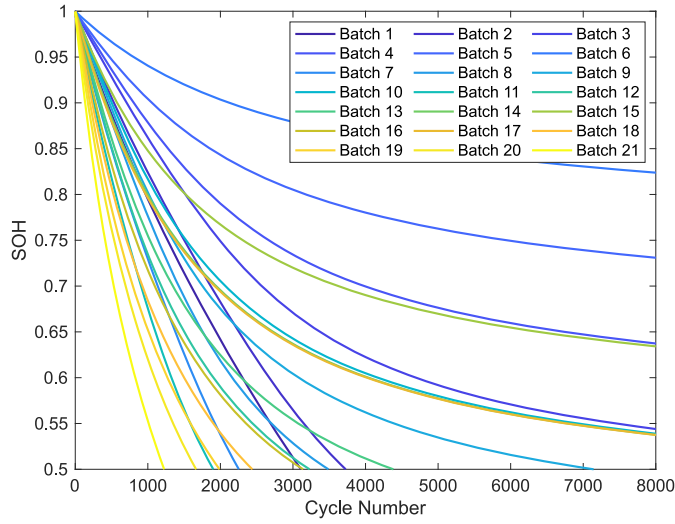


Fig. 5. SOH degradation trajectories of simulation data for 21 working conditions.

Table 2
Hyperparameter configuration of the calibrated PINN framework.

Component	Number of layers	Hidden dimensions
Bayesian SOH estimation network	20	100
Governing equation approximator network	8	72
Physics model calibrator network	12	72

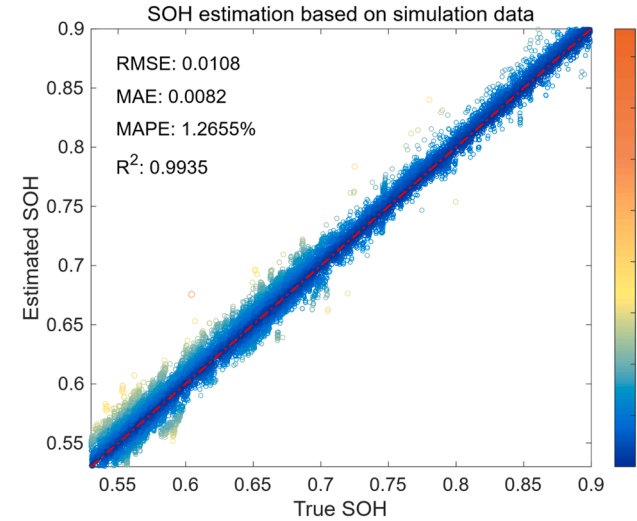


Fig. 6. SOH estimation results based on simulation data.

Table 3
Performance comparison of models based on real-world data.

Method	RMSE	MAE	MAPE	R ²
DNN	0.0166	0.0130	1.8522%	0.9715
Standard PINN	0.0159	0.0121	1.7369%	0.9757
Proposed method	0.0142	0.0098	1.4524%	0.9805

from the PET-based model. This simulation data serves as the foundation for the pre-training of the governing equation approximator. The approximator is initially trained using the simulation data to capture the

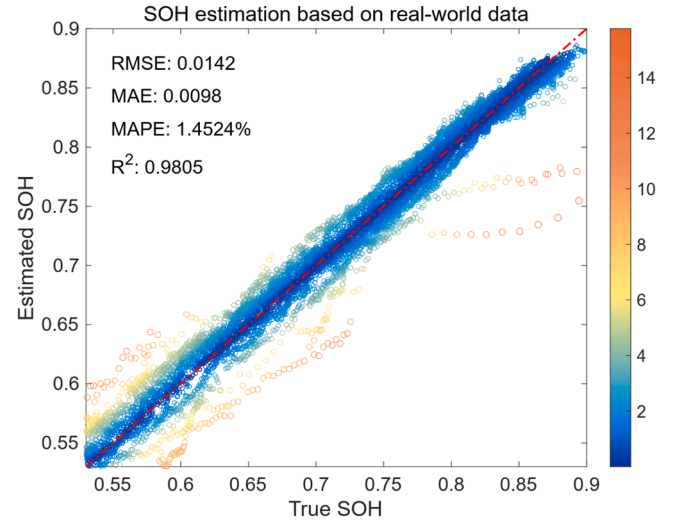


Fig. 7. SOH estimation results based on real-world data.

general degradation dynamics of the battery. Once pre-training is complete, the governing equation approximator is frozen, and Bayesian SOH estimation model is trained with real-world data, under the physics-informed constraints.

During both training and inference, MC Dropout is employed to estimate the uncertainty in the estimations by approximating the Bayesian posterior of network weights. This allows for predictive means and CIs to be generated. The total loss function is a combination of a data fidelity term and a calibrated physics-informed term. The data fidelity term ensures that the model aligns with real-world observations, while the calibrated physics-informed term incorporates the constraints from the governing equations, ensuring physical consistency in the estimations.

The training process iteratively updates the parameters of the Bayesian SOH estimator and the residual calibration DNN. This iterative optimization allows the framework to achieve accurate and reliable SOH estimations for SL-LIBs, even in the presence of sparse data and complex aging mechanisms. The complete training and optimization procedure is shown in Algorithm 1.

4. Case study

This section demonstrates the performance of the proposed calibrated PINN framework on a second-life battery dataset. The experimental conditions, feature extraction and visualization details, comparison of the calibrated PINN with baseline methods, and uncertainty quantification results are presented in this section.

4.1. Experimental setup

1) Feature generation

The dataset consists of 84 SL-LIBs sourced from retired EV packs, as described in Section 2.2. Features are extracted from seven key discharge curves, including voltage, current, capacity, energy, dQ/dV (the derivative of capacity with respect to voltage), temperature, and corresponding time. These curves provide critical information of battery behavior across different working conditions. For each curve, eleven feature extraction methods are employed, including mean, standard deviation, variance, range, maximum, minimum, skewness, kurtosis, and three quartiles (25%, 50%, and 75%). By applying eleven extraction methods to the seven curves, a total of 77 features are initially obtained.

To select the most informative features, we combine Pearson correlation scores and random forest importance metrics. The top 20 features are chosen, such as the mean of the dQ/dV curve, the quartiles 25% of

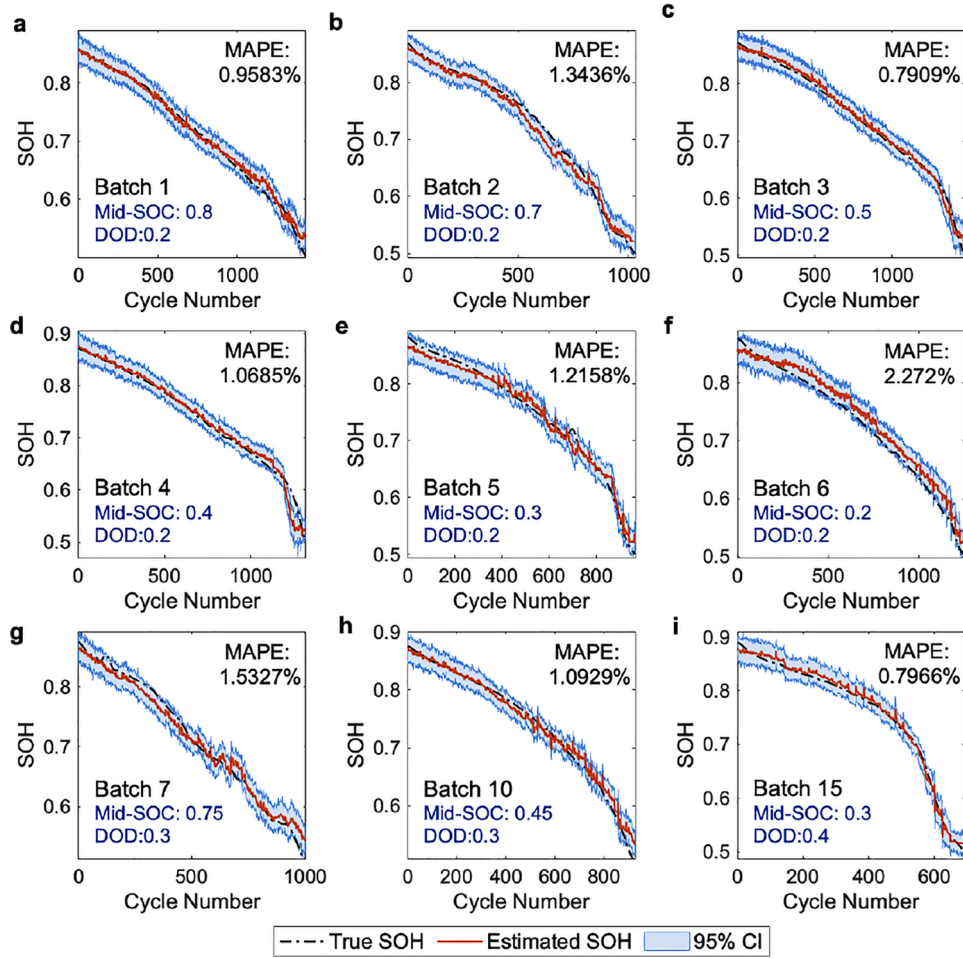


Fig. 8. SOH estimation results with 95% CIs for representative batches.

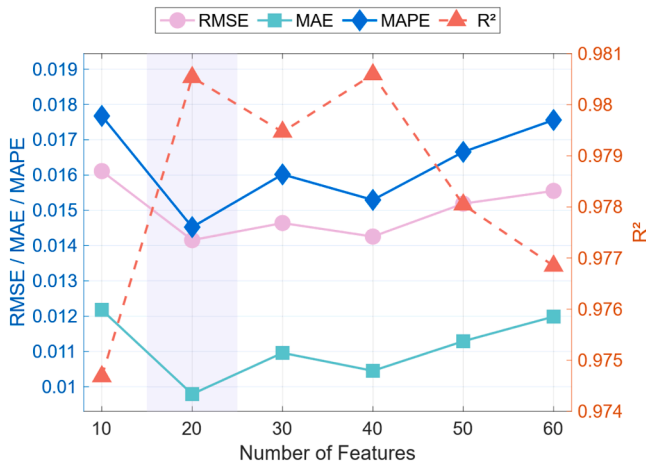


Fig. 9. Impact of number of features on model performance.

the dQ/dV curve, slope of voltage curve, and so on. The feature selection process reduces the computational complexity while retaining the most critical information for SOH estimation. Fig. 4 visualizes the top 12 of these selected features.

2) Simulation data generation

In addition to the real-world dataset, simulation data is generated

Table 4
Performance comparison with different numbers of features.

Feature number	RMSE	MAE	MAPE	R ²
10	0.0141	0.0098	1.4524%	0.9805
20	0.0142	0.0098	1.4524%	0.9805
30	0.0146	0.0110	1.6018%	0.9795
40	0.0143	0.0105	1.5294%	0.9806
50	0.0152	0.0113	1.6655%	0.9780
60	0.0156	0.0120	1.7558%	0.9768

using a PET-based battery degradation model [35] that captures the physical degradation processes of lithium-ion batteries. The simulation is designed to replicate the actual working conditions of the batteries, ensuring that the cycling profiles, charge/discharge rates, and temperature settings match those used in the experimental setup. The size of the simulated data in this study is determined based on the need to cover a sufficient range of real-world working conditions. This ensures that the simulated data closely aligns with the real-world data, providing a consistent basis for model training.

The simulation is conducted across 21 different working conditions, which mirror the 21 experimental conditions used in the real-world dataset. For each working condition, corresponding simulation data is generated, with the SOH degrading from 100% to 50%, covering both the first life and second life ranges. Since the starting capacity of second-life batteries can vary, the simulation generates data covering a range of initial capacities to capture the full range of second-life battery behavior. The SOH degradation trajectories for each working condition

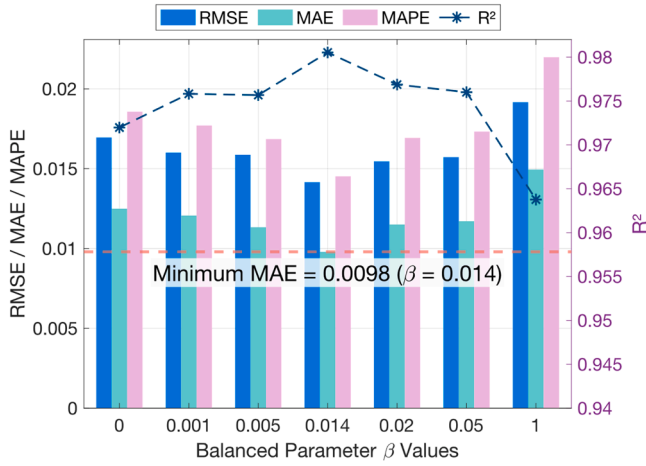


Fig. 10. Impact of the regularization parameter β on model performance.

Table 5

Performance comparison with different values of β .

β value	RMSE	MAE	MAPE	R^2
0	0.0170	0.0125	1.8575%	0.9720
0.001	0.0160	0.0121	1.7708%	0.9758
0.005	0.0159	0.0113	1.6857%	0.9757
0.014	0.0142	0.0098	1.4524%	0.9805
0.02	0.0155	0.0115	1.6933%	0.9769
0.05	0.0157	0.0117	1.7327%	0.9760
1	0.0192	0.0149	2.1987%	0.9638

Table 6

Performance comparison with published methods.

Method	RMSE	MAE	MAPE	R^2
MLP [23]	0.0241	0.0169	2.7838%	0.9591
GRU [48]	0.0269	0.0191	3.0747%	0.9514
LSTM [48]	0.0263	0.0188	3.1210%	0.9470
CNN [48]	0.0239	0.0183	2.8584%	0.9637
Attention [48]	0.0251	0.0180	2.8768%	0.9580
Proposed method	0.0142	0.0098	1.4524%	0.9805

shown in Fig. 5. The simulation data contains a total of 103717 sample points, from which 80% of the simulation samples is randomly selected for model training and remaining 20% is reserved for testing.

As in the feature generation process for real-world condition monitoring data, the same set of features is extracted from the simulation data. These features are then processed using the same feature extraction methods to ensure consistency between the real-world and simulated datasets. This simulation data, containing both the degradation dynamics and the same set of features as the real-world data, is used to pre-train the governing equation approximator in the calibrated PINN framework. The governing equation approximator is initialized with this simulation data, which helps the model to learn the expected battery behavior under various operational conditions before being fine-tuned with the real-world data.

3) Model configuration

The model structure used for training on both the simulation data and the real-world data follows the same architecture, employing fully connected neural networks for the three main components: the Bayesian SOH estimation network, the governing equation approximator network, and the physics model calibrator network. These components work together to estimate SOH, approximate the governing equations,

and calibrate discrepancies between the physics model and real-world data.

Each component has a specific number of layers and hidden dimensions to learn complex relationships effectively, as summarized in Table 2. The model is trained for 1000 epochs with a learning rate of 0.001. The balanced parameter β , which balances the loss terms, is set to 0.014. The dropout is set to 0.04, the Adam optimizer is used for parameter updates, and all hyperparameters are optimized using Bayesian optimization to ensure the best possible performance. The optimization process iteratively selects hyperparameter configurations using a Gaussian process model to predict performance, balancing exploration and exploitation for efficient optimization and optimal model performance on the validation set.

4.2. Experiment results

To evaluate SOH estimation performance, four metrics are employed: RMSE, mean absolute error (MAE), MAPE, and coefficient of determination (R^2), as these metrics are widely used in regression tasks and provide a comprehensive understanding of the model's accuracy. The formulas for these metrics are as follows: $RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (u_i - \hat{u}_i)^2}$, $MAE = \frac{1}{N} \sum_{i=1}^N |u_i - \hat{u}_i|$, $MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{u_i - \hat{u}_i}{u_i} \right| \times 100\%$, and $R^2 = 1 - \frac{\sum_{i=1}^N (u_i - \hat{u}_i)^2}{\sum_{i=1}^N (u_i - \bar{u})^2}$. Here, u_i represents the actual SOH, $\hat{u}_i$ represents the predicted SOH, and N is the number of observations.

The testing results for the simulation data are illustrated in Fig. 6, showing the SOH estimation performance of the proposed method. The fitting performance of the trained model based on the simulation data indicates that the governing equation approximator is well-trained, capturing the expected battery degradation behavior across different working conditions.

For the real-world dataset, each of the 21 batches consists of four batteries: three are used for training, and the remaining one is used for testing. Specifically, 10% of the training dataset is randomly selected for Bayesian optimization to search for optimal hyperparameters. Using the governing equation approximator trained on the simulation data, the model can capture degradation behavior from the physics model, ensuring that the SOH estimation process is grounded in physical principles. To provide benchmark comparisons, two baseline models are considered:

- 1) Purely data-driven model: A conventional DNN trained solely on input-output data, without incorporating any physical knowledge.
- 2) Standard PINN: This baseline model uses the same physical constraints and real-world data as the proposed method but does not include the physics model calibrator.

Table 3 summarizes the performance metrics, including RMSE, MAE, MAPE, and R^2 , for each of the models. The results shown in Table 3 demonstrate that the proposed method outperforms both the purely data-driven DNN model and the standard PINN in terms of predictive accuracy. Our model achieves a 16.3% reduction in MAPE compared to the standard PINN approach, demonstrating a significant enhancement in SOH estimation accuracy. Fig. 7 illustrates the SOH estimation results using the proposed method, showing the estimated values plotted against the true values. The red line represents the ground truth. Most estimations align closely with actual SOH values, indicating high accuracy.

To quantify uncertainty, a 95% confidence interval is constructed by drawing 100 MC Dropout samples and using the 2.5% and 97.5% percentiles of the estimations. The coverage probability of the 95% CI reaches 89.47%, indicating high reliability. The mean interval width of 0.0445 further highlights the framework's capacity to provide precise and informative uncertainty bounds.

Additionally, Fig. 8 shows the SOH estimation results for nine

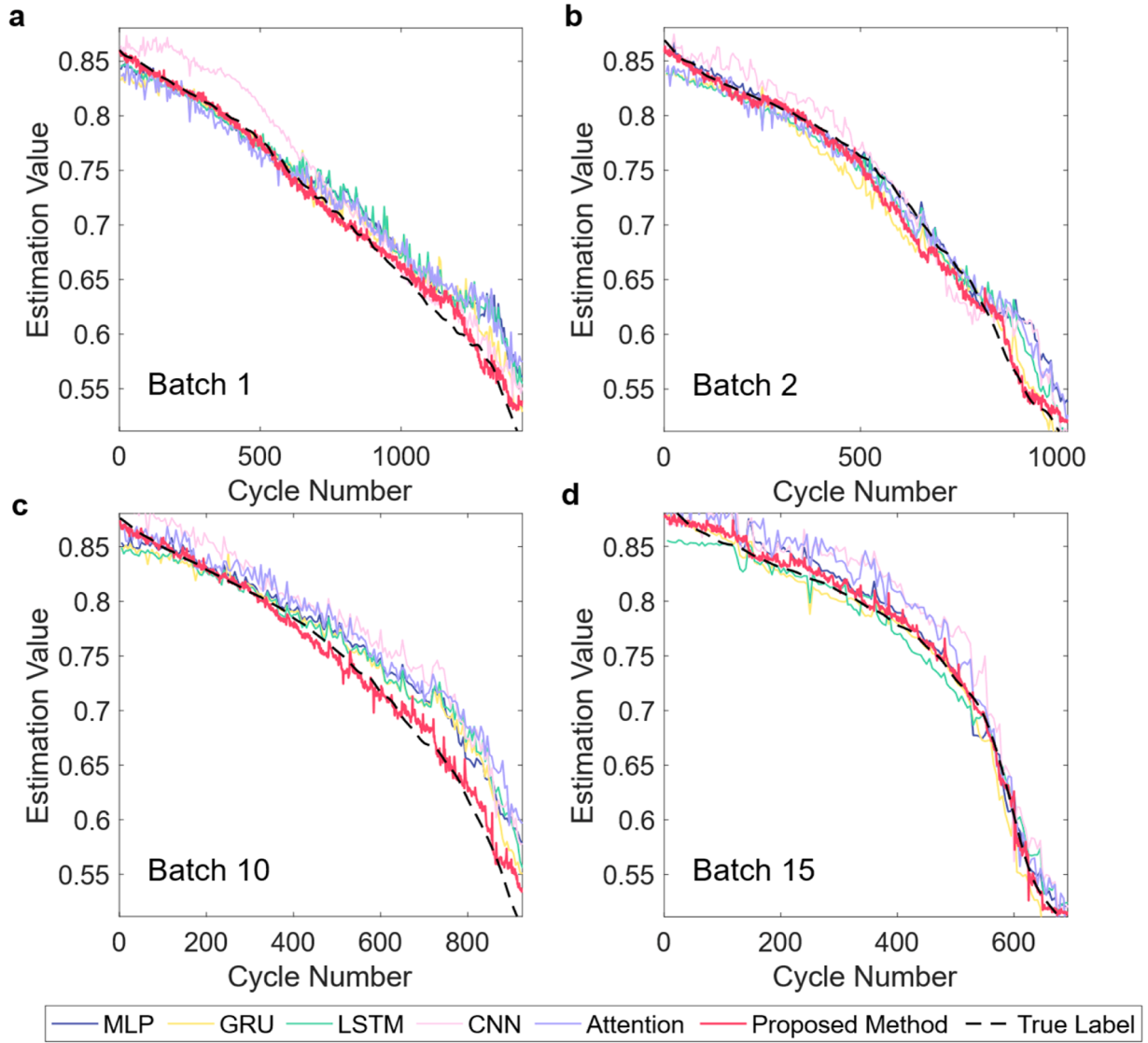


Fig. 11. Comparison between the proposed method and baseline models.

representative batches (i.e., Batch 1-7, Batch 10, and Batch 15). The solid red line represents the estimated SOH, and the dashed black line shows the ground truth. The shaded blue region represents the 95% CI, capturing the uncertainty in the estimations. For most cycles, the CIs remain narrow, reflecting robust uncertainty quantification. Although minor discrepancies occur in later cycles when SOH degradation accelerates, the estimation results closely follow the true SOH trend. These results demonstrate the effectiveness of the proposed framework in accurately capturing SOH trends under different working conditions.

4.3. Sensitivity analysis

1) Impact of feature numbers on model performance

Fig. 9 and Table 4 illustrates how the number of input features influences the performance of the proposed method, measured using RMSE, MAE, MAPE, and R^2 . The results demonstrate that the model's performance improves significantly as the number of features increases, reaching optimal metrics when approximately 20 features are used (shaded region in Fig. 9). Beyond this point, performance slightly degrades, suggesting that including additional features does not substantially enhance model performance and may introduce noise or overfitting.

These results validate the effectiveness of the feature selection process, confirming that selecting around 20 features provides the best trade-off between model complexity and predictive accuracy.

2) Impact of the regularization parameter β

The value of β is determined through Bayesian optimization to balance the contributions of the physics-informed and data-driven components. The sensitivity of the model to different settings of β is analyzed, which balances data fidelity and physics-informed constraints. To ensure consistency, the same dataset described in Section 2 is used, with the same train-test split as in previous experiments. Each candidate β value is tested in a complete training-evaluation cycle.

Fig. 10 and Table 5 illustrates the effect of varying β on the model's performance. As shown in Fig. 10 and Table 5, a small β (e.g. 0.001) favors data fitting but may overlook physical constraints, while a large β (e.g. 1) strictly enforces the physics term at the potential cost of data accuracy. The results indicate that $\beta=0.014$ provides optimal performance, achieving the lowest MAE of 0.0098 (orange dashed line in Fig. 10). Although the value of β may appear small, it plays an important role in balancing the contributions of the physics-informed and data-driven components. The small β value reflects the specific characteristics of the SL-LIBs dataset, where the real-world data provides strong

guidance. By tuning β , we can fine-tune the model to either prioritize data fitting or align more strongly with physical principles.

4.4. Comparison with published methods

To further validate the effectiveness of the proposed method, comparative experiments are conducted against state-of-the-art methods for SOH estimation of SL-LIBs under partial charging-discharging conditions. We adopt the same feature set used by Jiang et al. [23] to ensure a fair comparison. Additionally, implementations of gated recurrent unit (GRU), long short-term memory (LSTM) networks, and convolutional neural networks (CNN) follow the recent architectures reviewed in [48].

As shown in Table 6, our framework achieves the best results, indicating that integrating physics-informed constraints with data-driven calibrator significantly improves accuracy. Fig. 11 compares SOH estimation curves from the proposed method with those from baseline models for four representative batches (Batch 1, Batch 2, Batch 10, and Batch 15). The dashed black line indicates true SOH values, while the red line represents the proposed method's estimation results. Our approach closely follows the actual SOH trajectories, especially during later cycles when degradation accelerates. By contrast, the other models exhibit larger deviations in these challenging regions, illustrating the superior accuracy and resilience of the proposed solution across various operating conditions.

Overall, the calibrated PINN framework consistently outperforms baseline and published methods in accuracy while also providing reliable uncertainty estimates. By incorporating physics-informed principles, a residual calibration network, and Bayesian inference through MC Dropout, the proposed approach effectively addresses the complex challenges of second-life battery SOH estimation.

5. Conclusion

This paper presented a Bayesian calibrated PINN framework that combines physics-based degradation models with data-driven residual calibrator to address model discrepancies and effectively capture the complex degradation mechanisms in LIBs. By leveraging the governing equation approximator pre-training using simulation data and integrating MC Dropout for uncertainty quantification, the framework provides accurate SOH estimations and generates reliable CIs, enhancing decision-making under uncertainty. The use of Pearson correlation and random forest importance scores for feature selection ensured optimal predictive performance while maintaining computational efficiency. Benchmark comparisons against MLP, GRU, LSTM, CNN, and Attention-based models demonstrated the superior accuracy of the calibrated PINN, achieving lower RMSE, MAE, MAPE, and a higher R^2 value. Experimental validation across 21 working conditions confirmed the framework's generalizability and its potential for supporting sustainable energy storage with second-life LIBs.

Future work may explore integration with advanced electrochemical models and real-time adaptive learning. While a fixed set of simulated data was used, expanding the simulated dataset to include more underrepresented working conditions may facilitate knowledge transfer and improve model generalization in real-world applications.

CRedit authorship contribution statement

Rong Zhu: Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Jiawen Hu:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Weiwen Peng:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grants 72471251 and 72171037, in part by the Shenzhen Sustainable Development Science and Technology Special Project (Double Carbon Special Project, KCXST20221021111208019).

Data availability

The data that support the findings of this study are available from the corresponding author, Jiawen Hu, upon reasonable request.

References

- [1] Luderer G, Madeddu S, Merfort L, et al. Impact of declining renewable energy costs on electrification in low-emission scenarios. *Nat Energy* 2022;7:32–42. <https://doi.org/10.1038/s41560-021-00937-z>.
- [2] Li N, Wang M, Lei Y, Yang B, Li X, Si X. Remaining useful life prediction of lithium-ion battery with nonparametric degradation modeling and incomplete data. *Reliab Eng Syst Saf* 2025;256:110721. <https://doi.org/10.1016/j.res.2024.110721>.
- [3] Lopez FA, Lauinger D, Vuille F, et al. On the potential of vehicle-to-grid and second-life batteries to provide energy and material security. *Nat Commun* 2024;15:4179. <https://doi.org/10.1038/s41467-024-48554-0>.
- [4] Tao S, Ma R, Zhao Z, Ma G, Su L, Chang H, et al. Generative learning assisted state-of-health estimation for sustainable battery recycling with random retirement conditions. *Nat Commun* 2024;15(1):10154. <https://doi.org/10.1038/s41467-024-54454-0>.
- [5] Ma R, Tao S, Sun X, et al. Pathway decisions for reuse and recycling of retired lithium-ion batteries considering economic and environmental functions. *Nat Commun* 2024;15:7641. <https://doi.org/10.1038/s41467-024-52030-0>.
- [6] Gao S, Sun F, Zhao X, Li Y. Optimal warranty period design for new products subject to degradation and environmental shocks considering imperfect maintenance. *Reliab Eng Syst Saf* 2025;256:110710.
- [7] Luo Y, Zhao X, Liu B, He S. Condition-based maintenance policy for systems under dynamic environment. *Reliab Eng Syst Saf* 2024;246:110072.
- [8] Hu J, Chen P. Mission abort policy considering imperfect alarms and two abort options. *Nav Res Log* 2025. <https://doi.org/10.1002/nav.22260>.
- [9] Mo R, Zhou H, Yin H, Si X. A survey on few-shot learning for remaining useful life prediction. *Reliab Eng Syst Saf* 2025;257:110850. <https://doi.org/10.1016/j.res.2025.110850>.
- [10] Demirci O, Taskin S, Schaltz E, Demirci BA. Review of battery state estimation methods for electric vehicles—Part II: SOH estimation. *J Energy Storage* 2024;96:112703. <https://doi.org/10.1016/j.est.2024.112703>.
- [11] Natella D, Onori S, Vasca F. A co-estimation framework for state of charge and parameters of lithium-ion battery with robustness to aging and usage conditions. *IEEE Trans Ind Electron* 2023;70(6):5760–70. <https://doi.org/10.1109/TIE.2022.3194576>. Jun.
- [12] Gao Y, Liu K, Zhu C, Zhang X, Zhang D. Co-estimation of state-of-charge and state-of-health for lithium-ion batteries using an enhanced electrochemical model. *IEEE Trans Ind Electron* 2022;69(3):2684–96. <https://doi.org/10.1109/TIE.2021.3066946>. Mar.
- [13] Xu A, Fang G, Zhuang L, Gu C. A multivariate student-t process model for dependent tail-weighted degradation data. *IIEE Trans* 2024. <https://doi.org/10.1080/24725854.2024.2389538>.
- [14] Baghdadi I, Briat O, Deléage JY, Gyan P, Vinassa JM. Lithium battery aging model based on Dakin's degradation approach. *J Power Sources* 2016;325:273–85. <https://doi.org/10.1016/j.jpowsour.2016.06.036>.
- [15] Xu X, Tang S, Yu C, Xie J, Han X, Ouyang M. Remaining useful life prediction of lithium-ion batteries based on Wiener process under time-varying temperature condition. *Reliab Eng Syst Saf* 2021;214:107675. <https://doi.org/10.1016/j.res.2021.107675>.
- [16] Zhou Z, Duan B, Kang Y, Shang Y, Zhang Q, Zhang C. Practical state of health estimation for LiFePO₄ batteries based on gaussian mixture regression and incremental capacity analysis. *IEEE Trans Ind Electron* 2022;70(3):2576–85. <https://doi.org/10.1109/TIE.2022.3167142>.
- [17] Zhang Y, Tang Q, Zhang Y, Wang J, Stimming U, Lee AA. Identifying degradation patterns of lithium-ion batteries from impedance spectroscopy using machine learning. *Nat Commun* 2020;11(1):1706. <https://doi.org/10.1038/s41467-020-15235-7>.
- [18] Lin M, Ke L, Meng J, Wang W, Wu J, Wang F. Health status estimation of lithium-ion battery under arbitrary charging voltage information using ensemble learning framework. *Reliab Eng Syst Saf* 2025;256:110782. <https://doi.org/10.1016/j.res.2024.110782>.
- [19] Chen B, Zeng X, Liu C, Xu Y, Cao H. Health management of power batteries in low temperatures based on adaptive transfer enforcer framework. *Reliab Eng Syst Saf* 2025;254:110613. <https://doi.org/10.1016/j.res.2024.110613>.

- [20] Wang Z, Yang L, Fang X, Zhang H, Xie M. Image-based remaining useful life prediction through adaptation from simulation to experimental domain. *Reliab Eng Syst Saf* 2025;255:110668. <https://doi.org/10.1016/j.res.2024.110668>.
- [21] Kang W, Wang D, Jongbloed G, Hu J, Chen P. Robust transfer learning for battery lifetime prediction using early cycle data. *IEEE Trans Ind Inform* 2025. <https://doi.org/10.1109/TII.2025.3545079>.
- [22] Braco E, San Martín I, Sanchis P, Ursúa A, Stroe DI. State of health estimation of second-life lithium-ion batteries under real profile operation. *Appl Energy* 2022; 326:119992. <https://doi.org/10.1016/j.apenergy.2022.119992>.
- [23] Jiang Y, Ke Y, Yang F, Ji J, Peng W. State of health estimation for second-life lithium-ion batteries in energy storage system with partial charging-discharging workloads. *IEEE Trans Ind Electron* 2024;71(10):13178–88. <https://doi.org/10.1109/TIE.2023.3344825>.
- [24] Camboim MM, Moreira AC, de Fátima NCM, Beck RF, Arioli VT, Omae C, Ding H. State of health estimation of second-life batteries through electrochemical impedance spectroscopy and dimensionality reduction. *J Energy Storage* 2024;78: 110063. <https://doi.org/10.1016/j.est.2023.110063>.
- [25] Wang Y, Huang H, Wang H, Wu X. A fast estimation method for state-of-health of retired batteries based on health features. *J Energy Storage* 2023;72:108677. <https://doi.org/10.1016/j.est.2023.108677>.
- [26] Raissi M, Perdikaris P, Karniadakis GE. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J Comput Phys* 2019;378:686–707. <https://doi.org/10.1016/j.jcp.2018.10.045>. Feb.
- [27] Jiang C, Zhong T, Choi H, Youn BD. Physics-informed Gaussian process probabilistic modeling with multi-source data for prognostics of degradation processes. *Reliab Eng Syst Saf* 2025;258:110893. <https://doi.org/10.1016/j.res.2025.110893>.
- [28] Wen P, Ye ZS, Li Y, Chen S, Xie P, Zhao S. Physics-informed neural networks for prognostics and health management of lithium-ion batteries. *IEEE Trans Intell Veh* 2024;9(1):2276–89. <https://doi.org/10.1109/TIV.2023.3315548>. Jan.
- [29] Wang F, Zhai Z, Zhao Z, Di Y, Chen X. Physics-informed neural network for lithium-ion battery degradation stable modeling and prognosis. *Nat Commun* 2024;15(1): 4332. <https://doi.org/10.1038/s41467-024-48779-z>.
- [30] Hofmann T, Hamar J, Rogge M, Zoerr C, Erhard S, Schmidt JP. Physics-informed neural networks for state of health estimation in lithium-ion batteries. *J Electrochem Soc* 2023;170(9):090524.
- [31] Sun G, Liu Y, Liu X. A method for estimating lithium-ion battery state of health based on physics-informed machine learning. *J Power Sources* 2025;627:235767.
- [32] He Z, Wang S, Shi J, Liu D, Duan X, Shang Y. Physics-informed neural network supported wiener process for degradation modeling and reliability prediction. *Reliab Eng Syst Saf* 2025;258:110906. <https://doi.org/10.1016/j.res.2025.110906>.
- [33] Singh S, Ebongue YE, Rezaei S, Birke KP. Hybrid modeling of lithium-ion battery: physics-informed neural network for battery state estimation. *Batteries* 2023;9(6): 301. <https://doi.org/10.3390/batteries9060301>. Art. no.
- [34] Kohtz S, Xu Y, Zheng Z, Wang P. Physics-informed machine learning model for battery state of health prognostics using partial charging segments. *Mech Syst Signal Process* 2022;172:109002. <https://doi.org/10.1016/j.ymssp.2022.109002>.
- [35] Berliner MD, Cogswell DA, Bazant MZ, Braatz RD. Methods-PETLION: opensource software for millisecond-scale porous electrode theory-based lithium-ion battery simulations. *J Electrochem Soc* 2021;168:090504. <https://doi.org/10.1149/1945-7111/ac201c>.
- [36] Zou Z, Meng X, Karniadakis GE. Correcting model misspecification in physics-informed neural networks (PINNs). *J Comput Phys* 2024;505:112918. <https://doi.org/10.1016/j.jcp.2024.112918>.
- [37] Chen Z, Xiu D. On generalized residual network for deep learning of unknown dynamical systems. *J Comput Phys* 2021;438:110362. <https://doi.org/10.1016/j.jcp.2021.110362>.
- [38] Basora L, Viens A, Chao MA, Olive X. A benchmark on uncertainty quantification for deep learning prognostics. *Reliab Eng Syst Saf* 2025;253:110513. <https://doi.org/10.1016/j.res.2024.110513>.
- [39] Nemani V, Biggio L, Huan X, Hu Z, Fink O, et al. Uncertainty quantification in machine learning for engineering design and health prognostics: a tutorial. *Mech Syst Signal Process* 2023;205:110796. <https://doi.org/10.1016/j.ymssp.2023.110796>.
- [40] Lin YH, Li GH. A Bayesian deep learning framework for RUL prediction incorporating uncertainty quantification and calibration. *IEEE Trans Ind Inform* 2022;18(10):7274–84. <https://doi.org/10.1109/TII.2022.3156965>.
- [41] Zhu R, Chen Y, Peng W, Ye ZS. Bayesian deep-learning for RUL prediction: an active learning perspective. *Reliab Eng Syst Saf* 2022;228:108758. <https://doi.org/10.1016/j.res.2022.108758>.
- [42] Li F, Min Y, Zhang Y, Zhang Y, Zuo H, Bai F. State-of-health estimation method for fast-charging lithium-ion batteries based on stacking ensemble sparse gaussian process regression. *Reliab Eng Syst Saf* 2024;242:109787.
- [43] Xu A, Wang R, Weng X, Wu Q, Zhuang L. Strategic integration of adaptive sampling and ensemble techniques in federated learning for aircraft engine remaining useful life prediction. *Appl Soft Comput* 2025;175:113067.
- [44] Lin M, You Y, Wang W, Wu J. Battery health prognosis with gated recurrent unit neural networks and hidden Markov model considering uncertainty quantification. *Reliab Eng Syst Saf* 2023;230:108978.
- [45] Zhang Y, Zhang M, Liu C, Feng Z, Xu Y. Reliability enhancement of state of health assessment model of lithium-ion battery considering the uncertainty with quantile distribution of deep features. *Reliab Eng Syst Saf* 2024;245:110002.
- [46] Zhang X, Liu M, Liu Y. Battery state of health estimation using an AutoGluon-tabular model incorporating uncertainty quantification. *J Energy Storage* 2024; 101:113920.
- [47] Gal Y, Ghahramani Z. Dropout as a bayesian approximation: representing model uncertainty in deep learning. In: *Proceedings of the international conference on machine learning*; 2016. p. 1050–9.
- [48] Wang F, Zhai Z, Liu B, Zheng S, Zhao Z, Chen X. Open access dataset, code library and benchmarking deep learning approaches for state-of-health estimation of lithium-ion batteries. *J Energy Storage* 2024;77:109884. <https://doi.org/10.1016/j.est.2023.109884>.